

Curriculum Vitae

Tsung-Hsuan TSAI

Post-doctoral researcher

Faculty of Mathematics, University of Vienna

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Nationality: Taiwan

Born: 13th April 1996

RESEARCH INTERESTS

- Geometric group theory: hyperbolic groups, random groups, small cancellation theory, van Kampen diagrams
- Topological dynamics: uniformly recurrent subgroups, topological groupoids

POSITIONS

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|-------------|--|
| 2026 - 2029 | University of Vienna, Austria
Postdoctoral researcher, with Goulmira Arzhantseva
Funded by FWF ESPRIT Program |
| 2025 - 2026 | Ashoka University, New Delhi, India
Postdoctoral researcher, with Suraj Krishna and Pritam Ghosh
Funded by Axis Bank Post-doctoral Program |
| 2023 - 2025 | Institut Camille Jordan, Lyon, France
Postdoctoral researcher, with Nicolás Matte Bon
Funded by LabEx MiLyon |
| 2022 - 2023 | Career interruption (military service, Taiwan) |
| 2019 - 2022 | Institut de Recherche Mathématique Avancée, Strasbourg, France
Ph.D. in Mathematics, supervised by Thomas Delzant
Funded by CDSN (Contrat doctoral spécifique normalien) |

EDUCATION

- | | |
|-------------|---|
| 2019 - 2022 | Université de Strasbourg, France
Ph.D. in Mathematics, defended July 2022
Supervisor: Thomas Delzant |
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Thesis: *Phase transitions in random group: free subgroups and van Kampen 2-complexes*

- 2016 - 2019 **École normale supérieure de Rennes, France**
Normalien à titre étranger
M.Sc. in Mathematics at Université de Rennes 1, 2019
B.Sc. in Computer Science at Université de Rennes 1, 2018
B.Sc. in Mathematics at Université de Rennes 1, 2017
- 2014 - 2016 **Lycée Janson de Sailly, Paris, France**
Classe préparatoire aux grandes écoles, MPSI and MP

RESEARCH INTERNSHIPS

- 05 - 06 2019 **Institut de Recherche Mathématique Avancée, Strasbourg, France**
Research internship M2, supervised by **Thomas Delzant**
Master thesis: *Random groups: the binomial triangular model*
- 06 - 07 2018 **Università di Padova, Padova, Italy**
Research internship M1, supervised by **Remke Kloosterman**
- 05 - 06 2017 **Laboratoire de Mathématiques Jean-Leray, Nantes, France**
Research internship L3, supervised by **François Laudenbach**

TEACHING

- 2025 - 2026 **Ashoka University, India**
Course Instructor, 36 hrs
Spring 2026: **Introduction to Geometric Group Theory**
Elective course MAT-4112 for 4th year students, developed and taught.
- 2019 - 2022 **Université de Strasbourg, France**
Teaching Assistant, 195 hrs
Autumn 2019, 2020, 2021: **TD Algèbre S5**
Introduction to group theory, exercise sessions for 3rd year math students, 99 hrs.
Spring 2020, 2021, 2022: **CI Math pour les sciences 2**
Basic linear algebra, lecturer for 1st year physics and chemistry students, 96 hrs.

PUBLICATIONS

- [1] T.-H. Tsai, *Density of random subsets and the intersection formula*.
In **Journal of Combinatorial Algebra**, volume 6, number 3/4, pp. 223–263 (2022).

DOI: [10.4171/JCA/63](https://doi.org/10.4171/JCA/63). ArXiv: [2104.09192](https://arxiv.org/abs/2104.09192).

- [2] T.-H. Tsai, *Freiheitssatz and phase transitions for the density model of random groups*.
In **Mathematische Zeitschrift**, 303, 65 (2023).
DOI: [10.1007/s00209-022-03186-2](https://doi.org/10.1007/s00209-022-03186-2). ArXiv: [2111.08958](https://arxiv.org/abs/2111.08958).
- [3] T.-H. Tsai, *Phase transition for the existence of van Kampen 2-complexes in random groups*.
In **Algebraic and Geometric Topology**, volume 24, issue 7, pp. 3897–3917 (2024).
DOI: [10.2140/agt.2024.24.3897](https://doi.org/10.2140/agt.2024.24.3897). ArXiv: [2210.08234](https://arxiv.org/abs/2210.08234).
- [4] T.-H. Tsai, *Parallel geodesics and minimal stable length of random groups*.
In **International Journal of Algebra and Computation** Vol. 35, No. 08, pp. 1135–1158 (2025).
DOI: [10.1142/S0218196725500377](https://doi.org/10.1142/S0218196725500377). ArXiv: [2410.07859](https://arxiv.org/abs/2410.07859)

SELECTED TALKS

- 12.2025 **Séminaire virtuel francophone Groupes et Géométrie (online)**
Van Kampen diagrams and parallel geodesics in random groups
- 11.2025 **Mathematics Colloquium, Ashoka University, New Delhi, India**
Geometry in random groups
- 03.2025 **Analysis and geometry seminar, University of Bristol, UK**
Parallel geodesics and minimal stable lengths in random groups
- 02.2024 **GADT seminar, IMB, Dijon, France**
Existence of van Kampen diagrams in random groups
- 10.2023 **Groupe de travail "Action!", ENS Lyon, France**
Introduction to random groups II: The Freiheitssatz
Introduction to random groups I: The density model
- 12.2021 **ENS Group theory seminar, Paris, France (Online)**
Freiheitssatz for the density model of random groups
- 10.2020 **Séminaire Doctorant, IRMA, Strasbourg, France**
Most groups are hyperbolic

LANGUAGES

• English (fluent) • French (fluent) • Mandarin (native) • Taiwanese (native)